

INTRODUCTION

The Amazon Sales Report dataset is a transactional dataset containing order details such as:

- Order ID
- Date
- Product Category
- Quantity
- Amount
- City/State
- Shipping Status
- Fulfillment Method
- Customer Information

This dataset can be used for multiple machine learning tasks including:

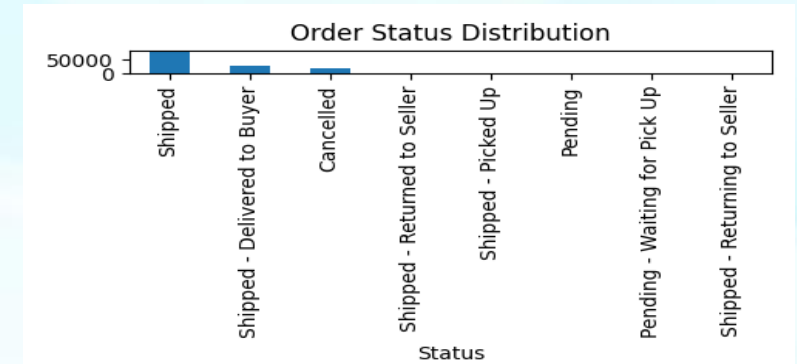
- Classification
- Regression
- Clustering
- Time Series Forecasting
- Recommendation Systems
- Anomaly Detection

EXPLORATORY DATA ANALYSIS [EDA]

- Observations – Amazon transactional records
- Variables – Sales amount, quantity, category, status, city etc.
- No major missing values after preprocessing

Summary:

- Data cleaning and preprocessing completed
- Sales trends analyzed
- Distribution of quantity and order status explored



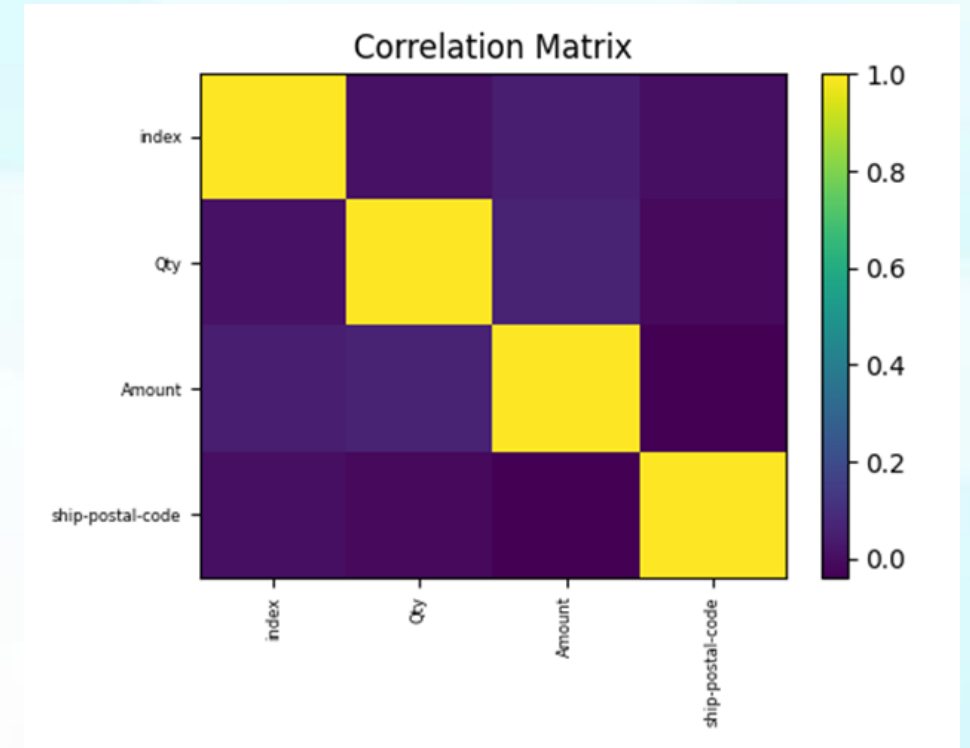
EXPLORATORY DATA ANALYSIS [EDA]

Correlation Analysis:

- Correlation close to +1 indicates strong positive relationship
- Correlation close to -1 indicates strong negative relationship
- High correlation may create multicollinearity

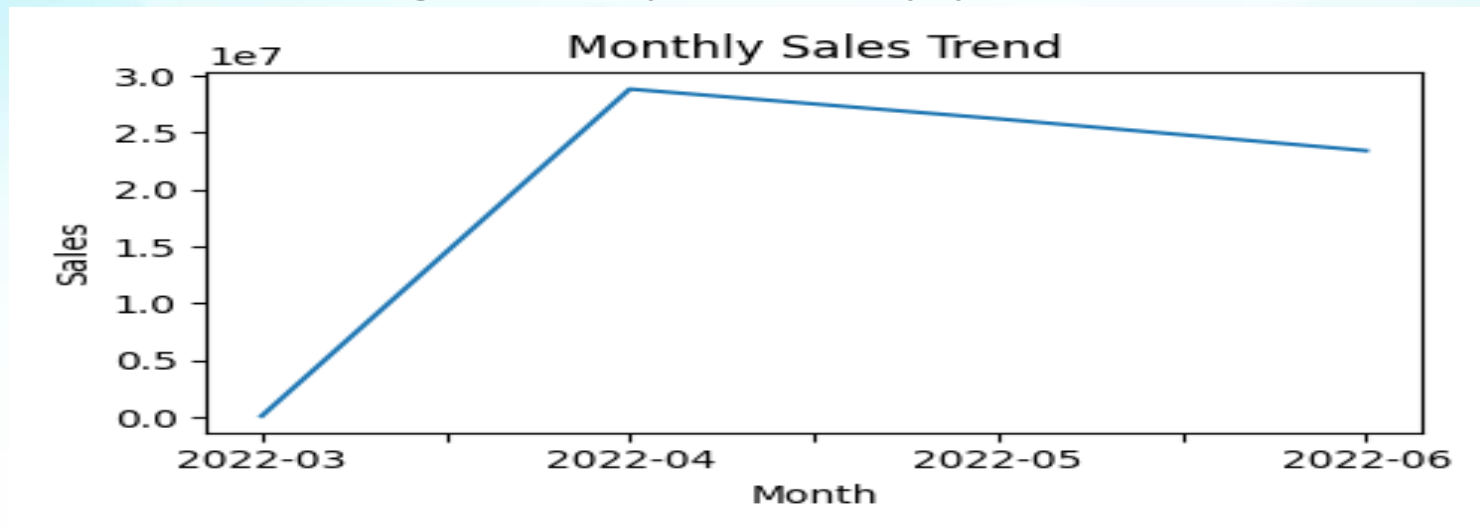
Checkpoint:

70% Train Data and 30% Test Data used for predictive modeling



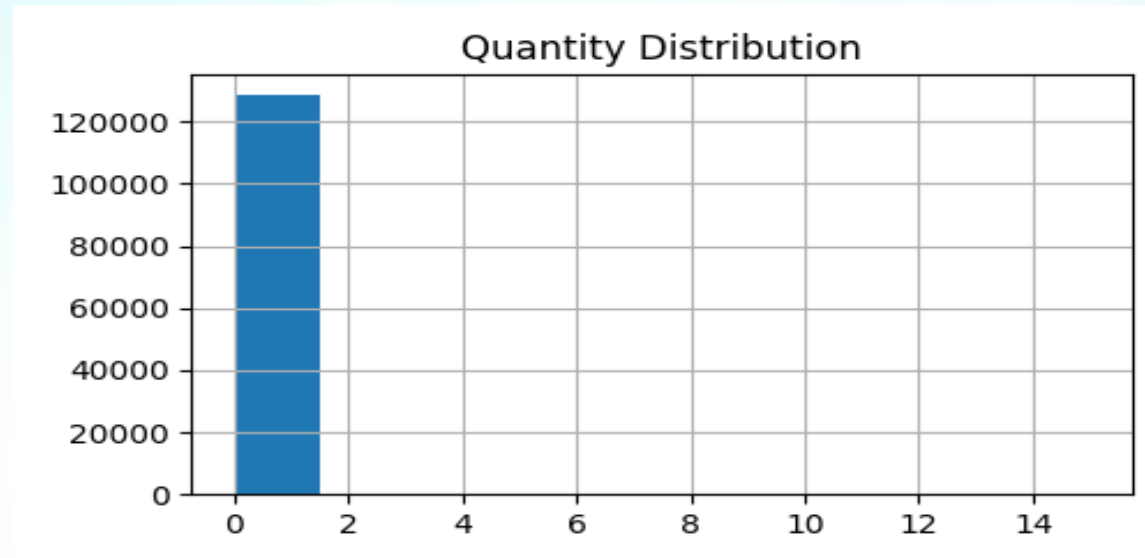
FULL MULTIPLE LINEAR REGRESSION MODEL

- Full model uses all predictors to estimate sales amount.
- Key Observations:
 - Multiple predictors improve prediction capability
 - Relationship between quantity, category and amount observed
 - Model helps identify important business drivers
- Adjusted R^2 indicates good explanatory power.



RIDGE REGRESSION FOR MODEL

- Ridge regression reduces overfitting and multicollinearity.
- Key Features:
 - Uses L2 regularization
 - Shrinks coefficients toward zero
 - Keeps all predictors in the model
- Cross-validation used to select optimal lambda value.



LASSO REGRESSION FOR MODEL

- LASSO performs variable selection and coefficient shrinkage.
- Key Features:
 - Uses L1 regularization
 - Some coefficients become zero
 - Produces simpler and interpretable models
- Cross-validation used to determine tuning parameter.

SUBSET SELECTION FOR MODEL

- Forward Stepwise Selection used for subset generation.
- Selection Criteria:
 - Adjusted R^2
 - BIC
 - Mallows C_p
- Best subset model selects only important predictors
- while reducing unnecessary variables.

CONCLUSION

- Amazon sales dataset supports predictive analytics.
- Multiple Linear Regression provides baseline understanding.
- Ridge and Lasso improve model stability.
- Best Subset Selection identifies important predictors.
- FINDINGS:
 - Quantity and category strongly influence sales.
 - Feature selection improves interpretability.
 - Regularization